Course Name: Object Oriented Concepts & Programming

Course Code: MCA113

Objectives:

Students will be introduced to

- Differentiate between procedural and object oriented programming
- Learn C++ as a language and various features to implement Object oriented concepts
- Learn Object Oriented principles and their application using C++

Prerequisites:

Knowledge of C language, Programming concepts including algorithm building and logic

Contents:

1. **Introduction to C++**
   Identifiers and constants (Literals), Keywords, Data Types, The Operators, New Casting Operators, Typeid and throw, The Conditional structures and Looping Constructs, The Difference between struct and class in C++, The difference between Union and Class, Static Data members of a class, Pointer to objects and pointer to members of class, The local classes, Assigning Objects

2. **Functions**
   Introduction, Inline function, Default Arguments to the function, Functions with object as parameters, Call by reference and return by reference, Prototyping and Overloading, Friend functions, Const and Volatile functions, Static functions, Private and Public functions, Function Pointers, Adding C functions to the C++ program

3. **Constructors and Destructors**
   Introduction to constructors, The explicit constructors, Parameterized constructors, Having multiple constructors, Constructors with default arguments, Dynamic Initialization, Constructor with dynamic allocation, Copy constructors, The member initialization list, Destructors

4. **Operator Overloading and User Defined Conversions**
   Introduction, Unary Operators, Binary Operators, Using Friends as operator functions, Overloading other Operators, The need for user defined conversion, Four different cases where user defined conversions are needed, Comparison of both the methods of conversion

5. **Templates**
   Function Templates, Non-Generic (Non-Type) Parameters in Template functions, Template function and specialization, Overloading a template function, Using Default Arguments, Class Templates, Classes with multiple generic data types, Static data members, Primary and Partial Specialization, The Export Keyword, The other use of typename

6. Inheritance

The need, Defining derived class using single base class, Derivation using public, private and protected access modifiers, The implementation of inheritance in the C++ object model, The Access Control, The Access Declaration, The multiple-inheritance, Abstract classes, Composite objects (container objects)

7. Runtime polymorphism by virtual functions

Compile Time and Runtime Polymorphism, Pointers to Objects, *this* pointer, Compatibility of Derived and base class pointers, The subobject concept, Virtual functions, Static invocation of virtual function, Default arguments to virtual functions, Virtual destructors, Pure virtual functions

8. IO Streams

Need for streams, Advantages of using C++ I/O over C IO, The C++ Predefined streams, Formatting IO, Formatting using *ios* members, Manipulators, Creating our own manipulator

9. Using Files for IO

Why IO is special, Text and binary streams, Opening and closing files, Dealing with text files Dealing with binary files, Providing Random Access using seek, IO Modes, Handling Errors

10. Namespaces

Introduction and need, Use the using syntax, Defining namespaces, Extending the namespace, Unnamed namespaces, Nested Namespaces, Namespace aliases, The *std* namespace, The Koenig lookup, Overhead with namespaces

11. The Standard Template Library

The STL (Standard Template Library) Introduction, Generic Programming, Generic Software Components, Generic Algorithms, Iterators, Containers, Algorithms

References:

1. Bhushan Trivedi, "*Programming with ANSI C++*", Oxford University Press
2. E. Balagurusamy, "*Object-Oriented Programming with C++*", McGraw Hill Publications
3. Marshall Cline, "*C++ FAQs*", Pearson Education
4. Stanley Lippmann, "*C++ Primer*", Pearson Education
5. Bjarne Stroustrup, "*The C++ Programming Language*", Pearson Education
6. Scott Mayer, "*Effective C++*", Addison-Wesley
7. Herbert Schildt, "*Complete Reference C++*", McGraw Hill Publications
8. ePgPathshala, "*Subject: Information Technology – P-01. Object Oriented Concepts and Programming using C++*", http://epgp.inflibnet.ac.in/Home/ViewSubject?catid=305

Accomplishments of the student after completing the course:

After completion of this course, students will be able to

- Understand and appreciate the Object Oriented approach of programming
- Get aware of the working and architectural model of C++
- solve problems given to him/her using C++ with keeping balance between efficiency and flexibility

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